Application of Artificial Intelligence, Machine Learning in Vehicle Analytics

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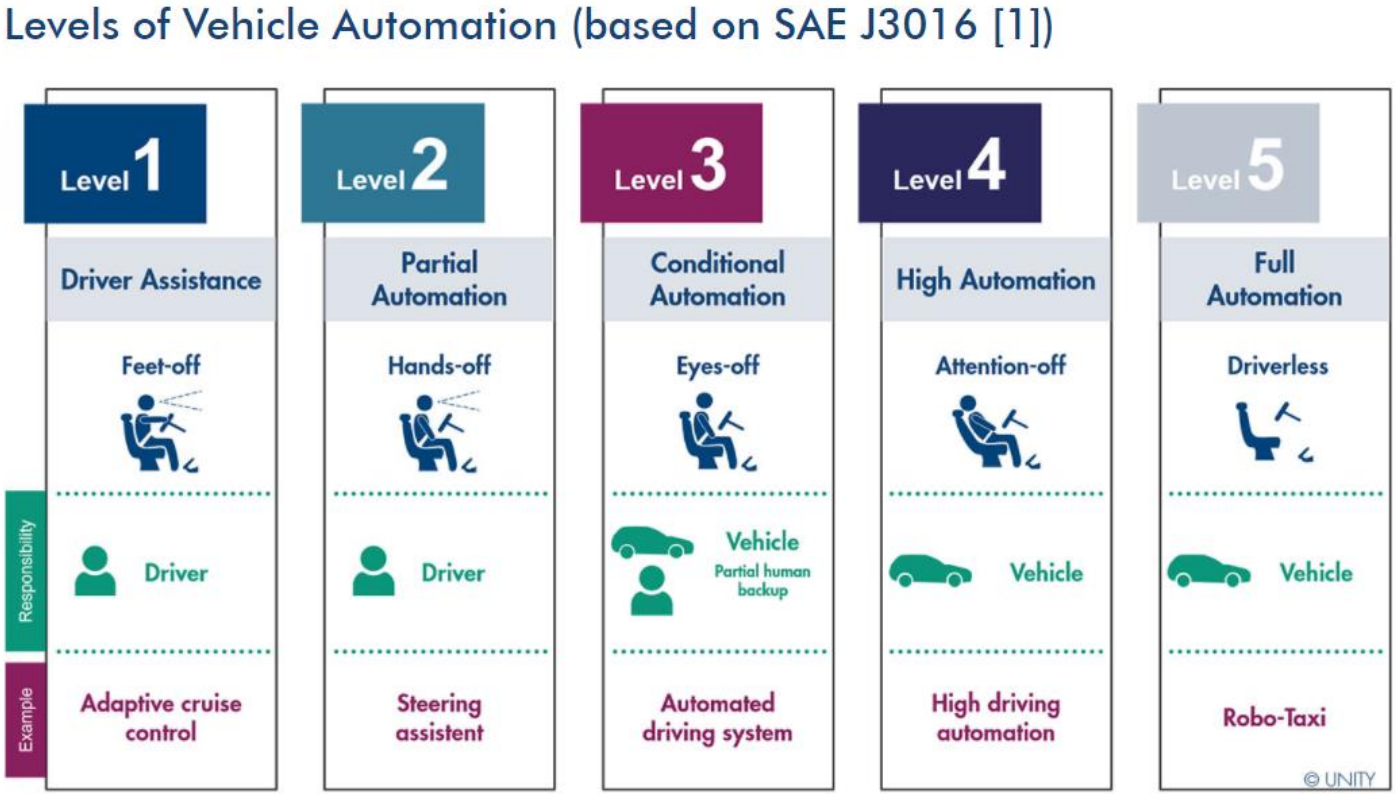
Agenda

Overview of :

- AI/Machine Learning/Deep Learning/Natural Language Processing
- Vehicle Analytics
- Smart Cockpit

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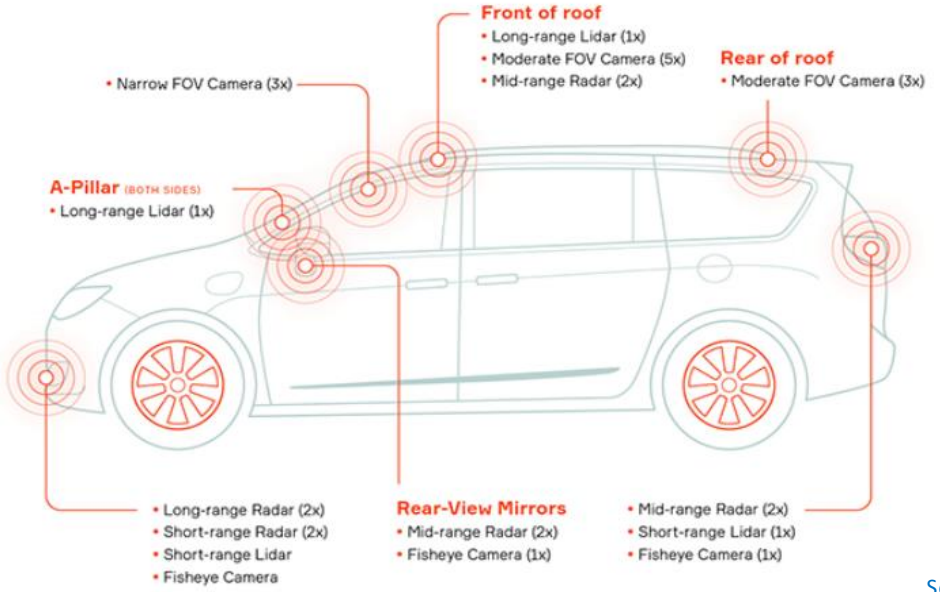
Smart Cockpit



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Smart Cockpit

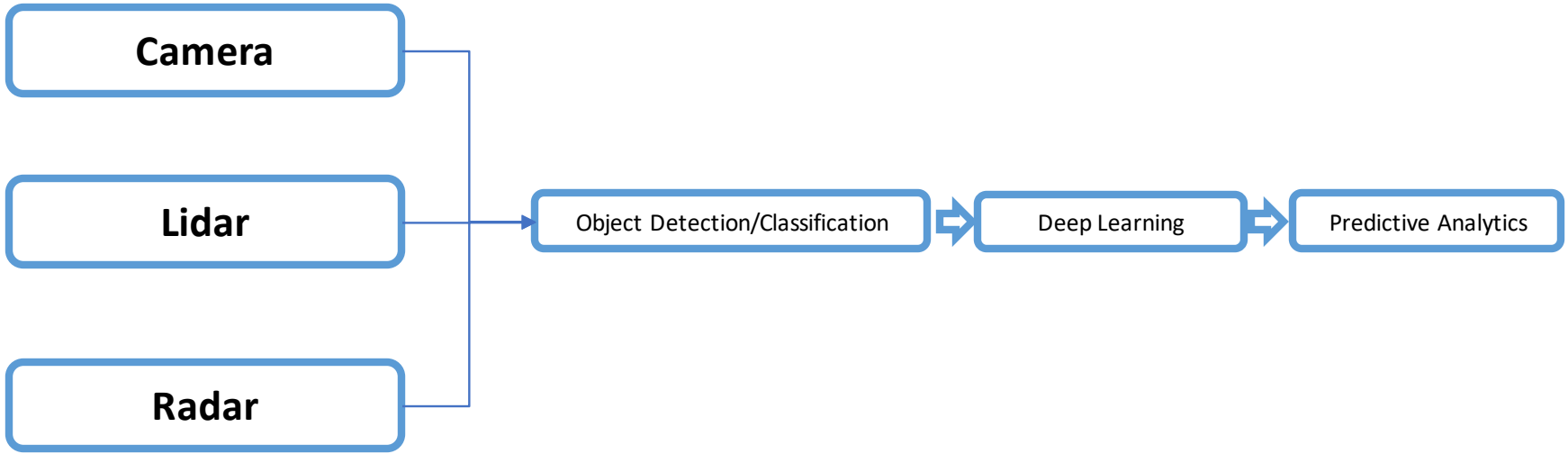
Sensors Enabling Level 4/5 Automated Vehicles



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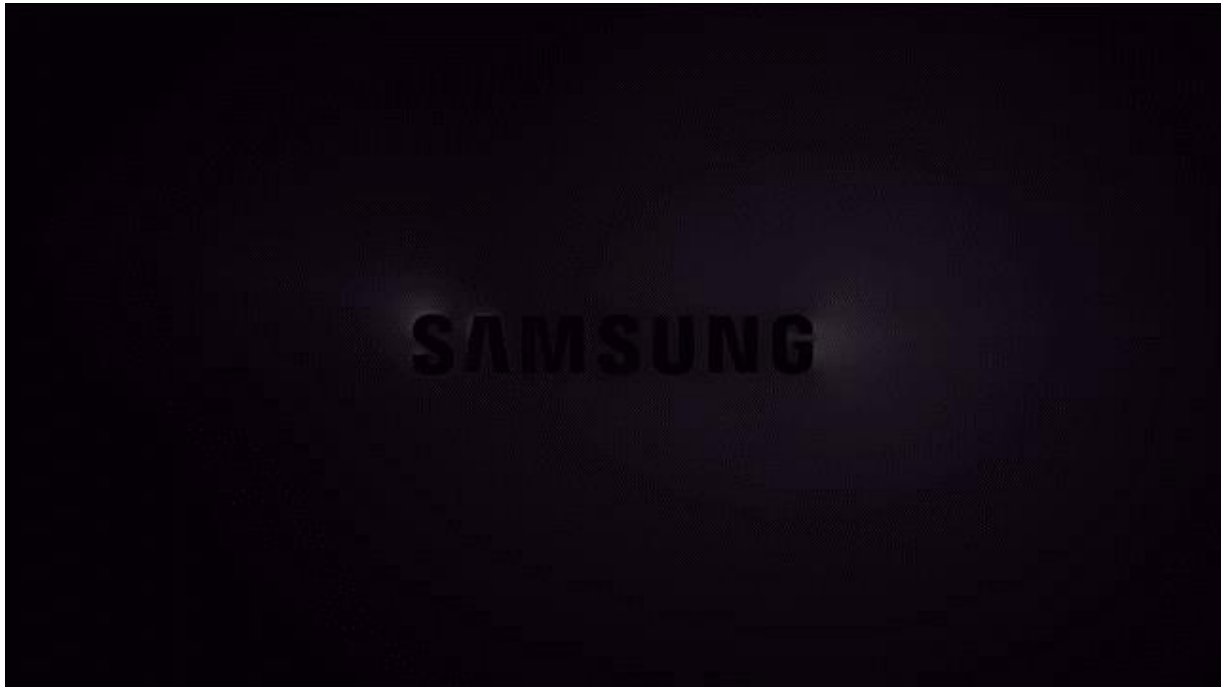
Smart Cockpit



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Smart Cockpit : Samsung Offering



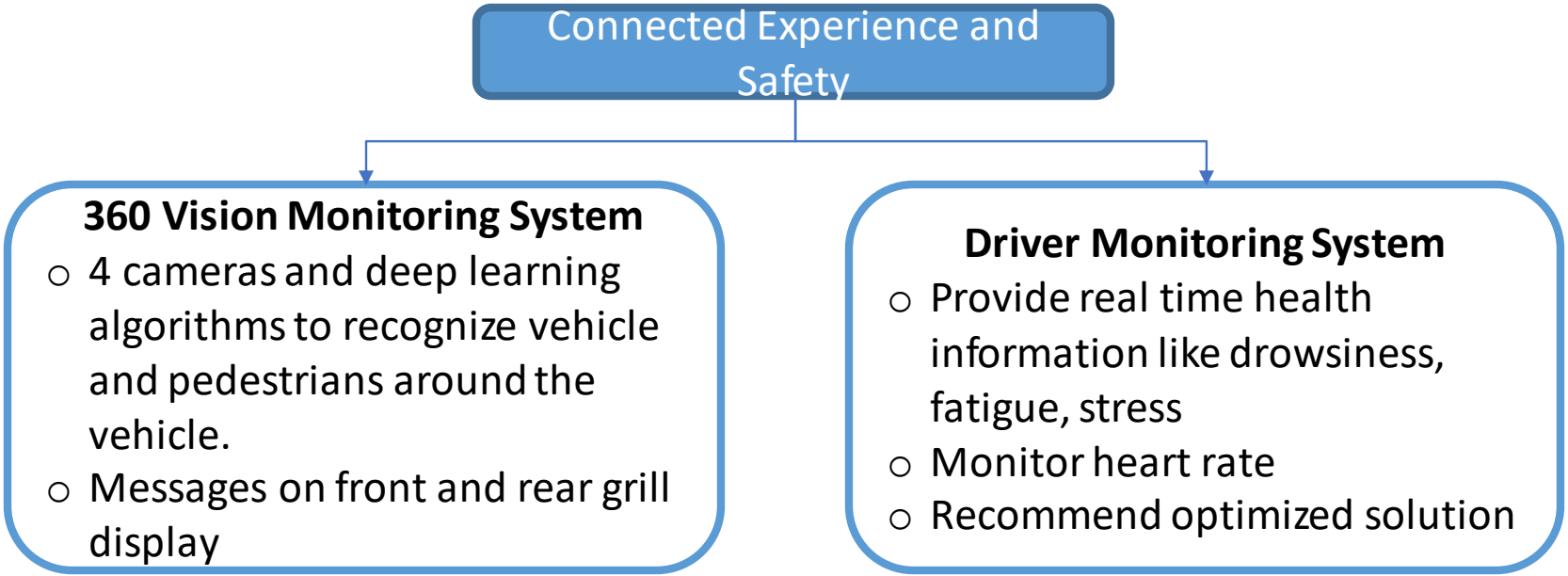
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Smart Cockpit : Samsung Offering



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Smart Cockpit : Samsung Offering

Connected Experience and Safety

360 Vision Monitoring System

- 4 cameras and deep learning algorithms to recognize vehicle and pedestrians around the vehicle.
- Messages on front and rear grill display

Image Classification → Deep Learning → Predictive Analytics

Recommendation System → Prescriptive Analytics

Driver Monitoring System

- Provide real time health information like drowsiness, fatigue, stress
- Monitor heart rate
- Recommend optimized solution

Predictive Analytics

Recommendation System → Prescriptive Analytics

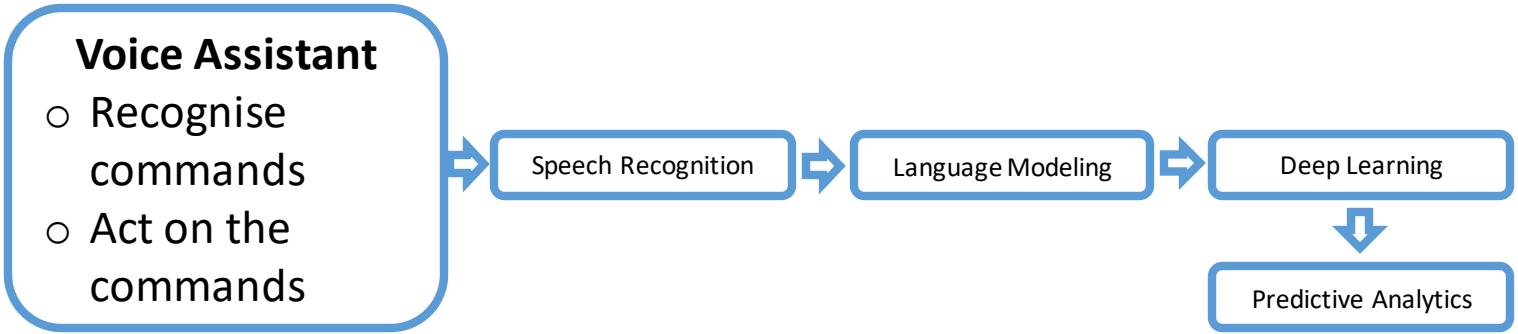
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Smart Cockpit : Voice Assistant



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Smart Cockpit : Voice Assistant



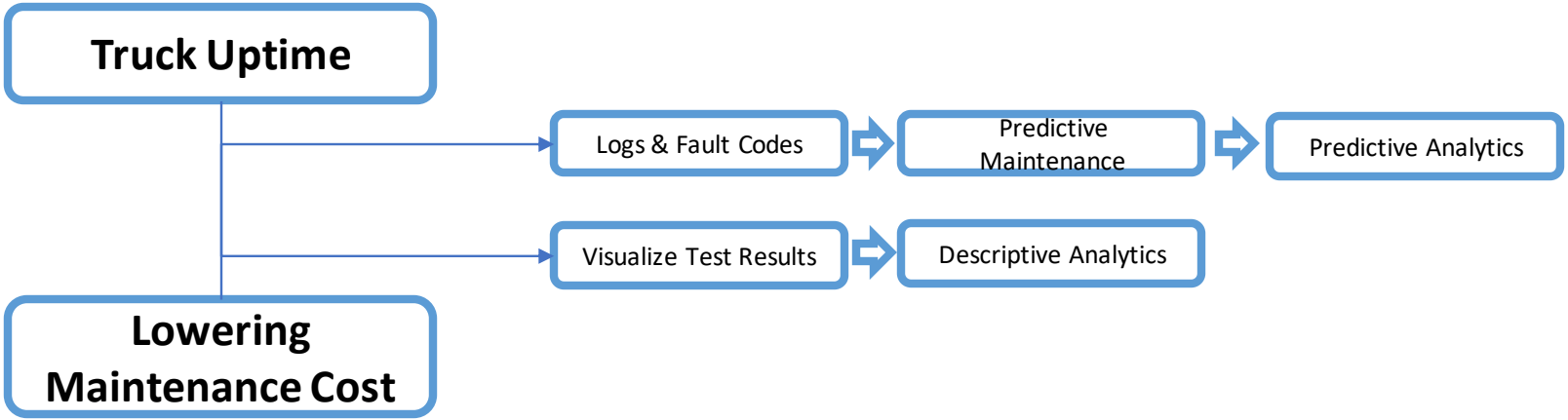
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Vehicle Analytics : KPIT Offering



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Vehicle Analytics : KPIT Offering



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Summary

Smart Cockpit

Vehicle Analytics

Image Recognition/ Classification

Speech Recognition

Recommendation Engine

Visualization

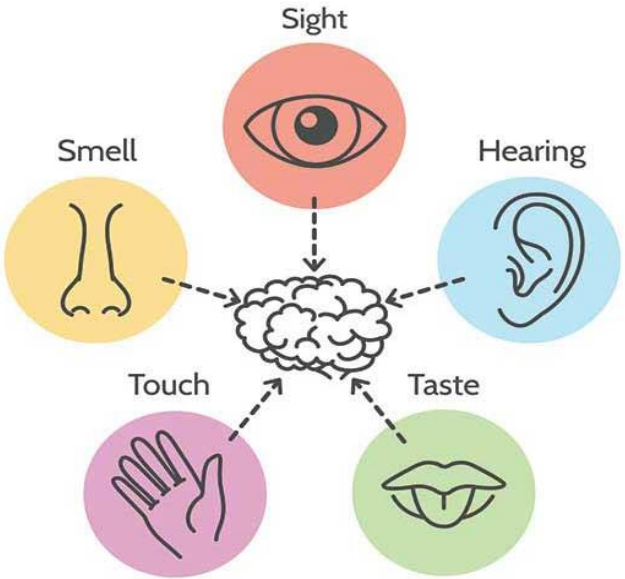
Artificial Intelligence

Machine Learning /Deep Learning

Descriptive/ Predictive/ Prescriptive Analytics

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What is intelligence?

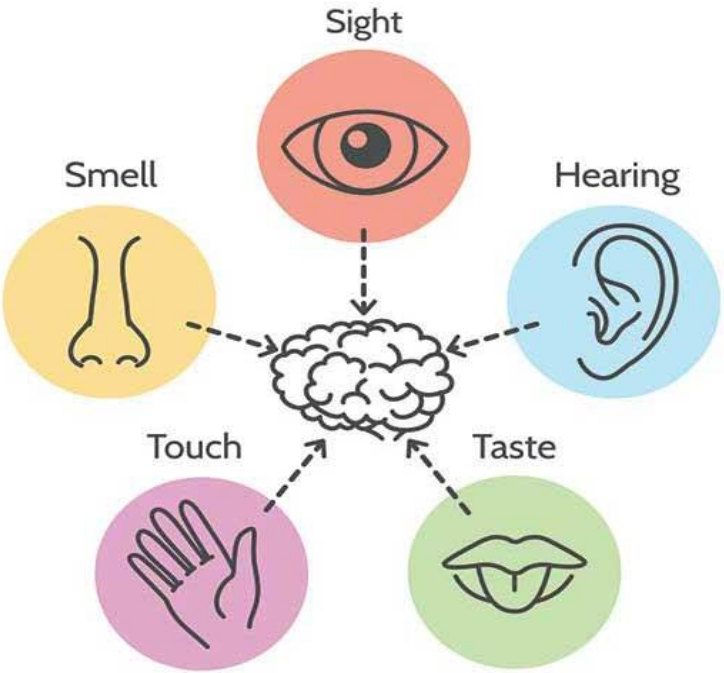


Some key skills we acquire:

- Object detection and recognition
- Emotion recognition through visual cues
- Speech
- Reading
- Understanding languages/music
- Understanding good touch from bad touch
- Decision making

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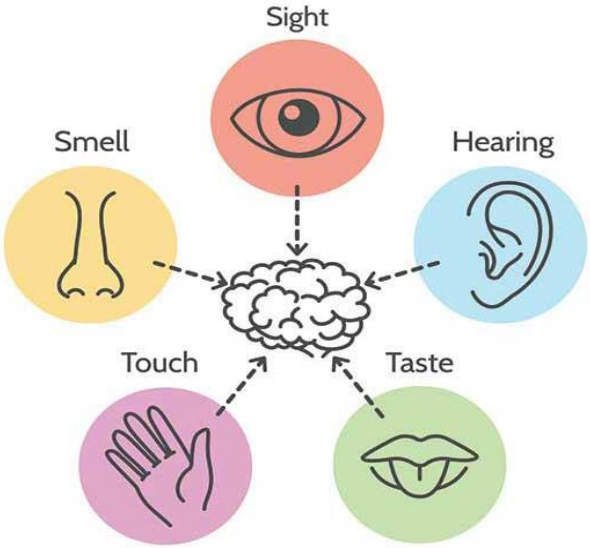
Driving Intelligence

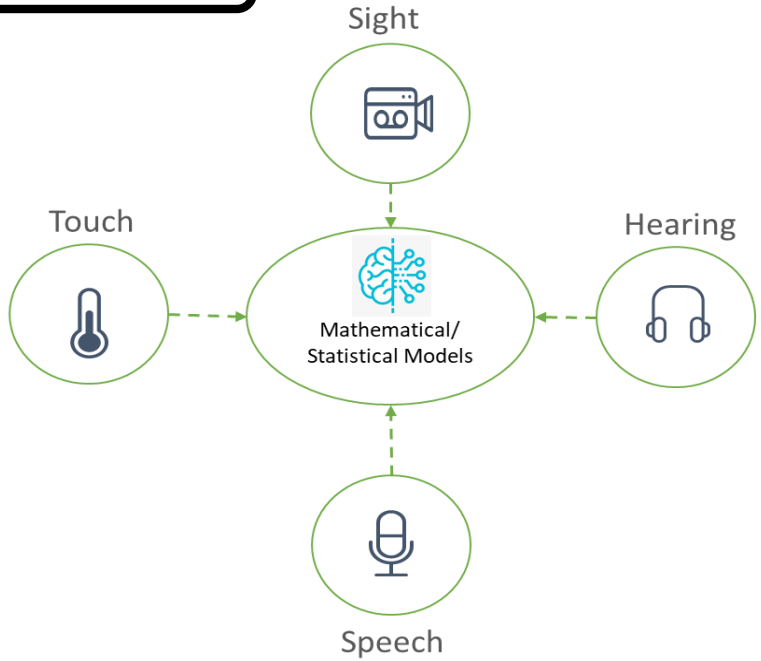


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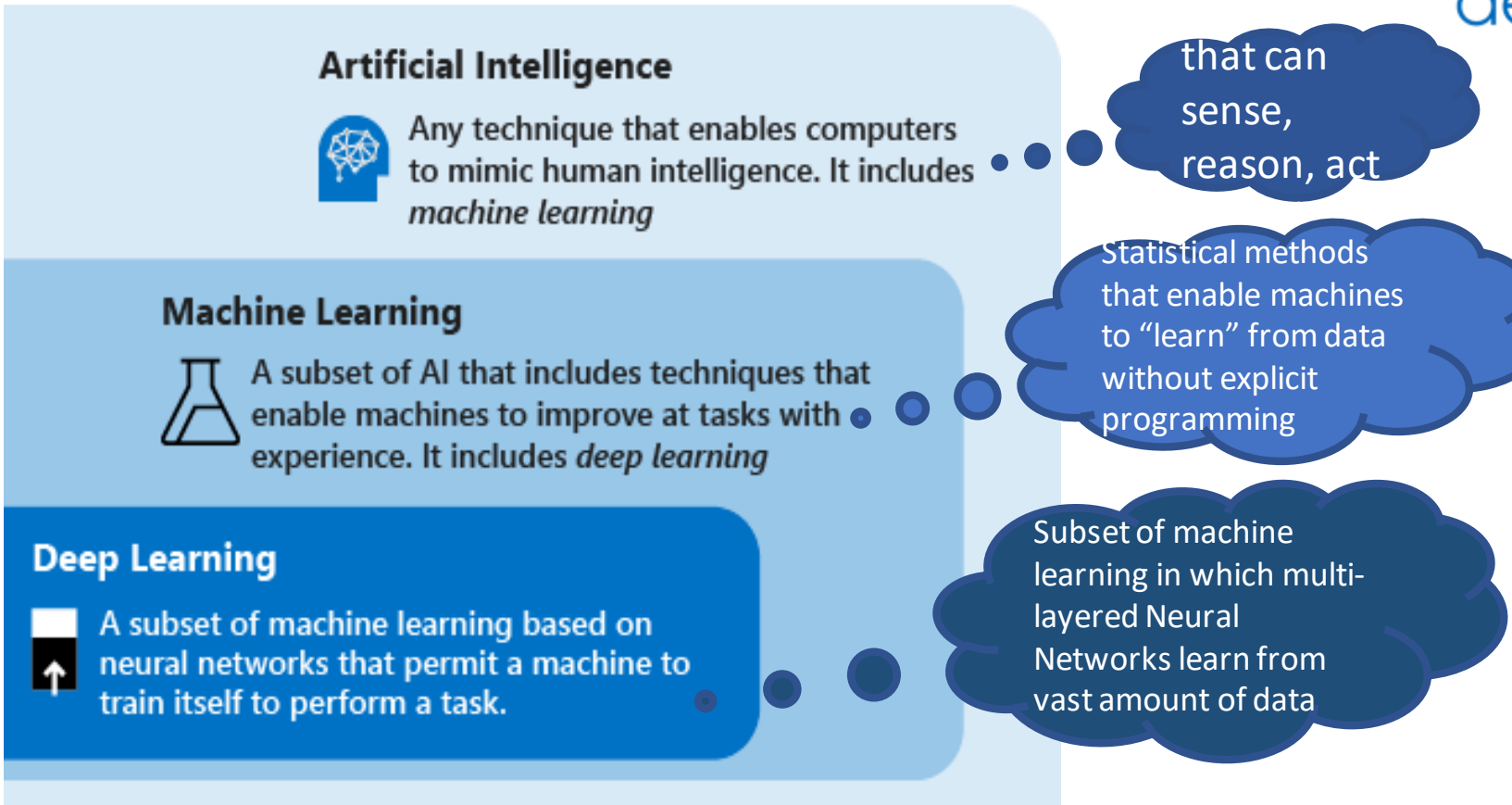
What is Artificial Intelligence?

Augmenting/Replacing human decision making





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Understanding Data Science

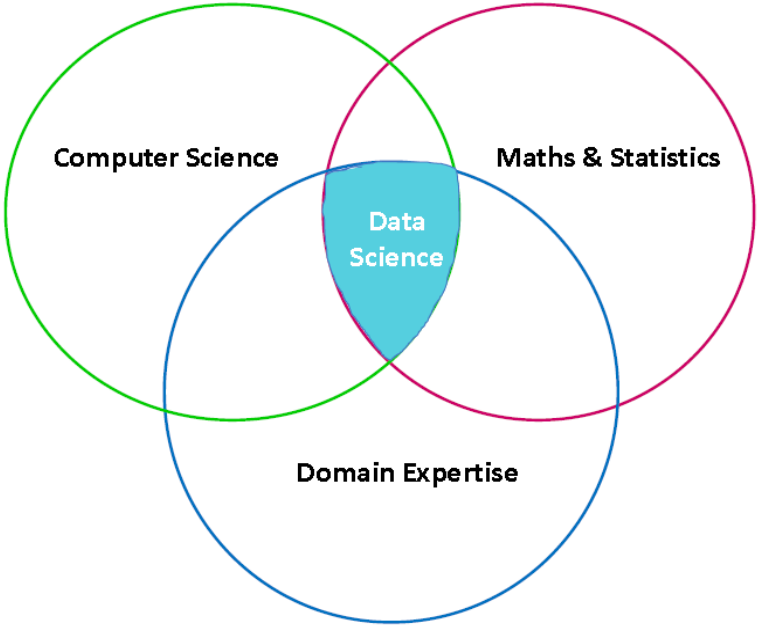


- Data Science refers to design, development & deployment of algorithms through statistical programming to support decision making.
- It includes:
  - Data acquisition/storage
  - Data cleansing
  - Data preparation
  - Model building
  - Model deployment

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Understanding Data Science

Data Science is a powerful combination of various disciplines.



Computer Science Skills:

- Programming
- Cloud technologies

Mathematics and Statistics Skills:

- Basic Statistics
- Derivatives
- Matrix algebra
- Machine Learning
- Deep Learning

Domain Knowledge:

- Banking/Finance/Insurance
- Retail
- Healthcare

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Cooking Process



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Data Science Methodology

Productionizing/Packaging

Schedule model runs to provide predictions at selected intervals. Monitor and tune models.

Data Analytics

Choose appropriate machine learning technique (Predictive Modeling, Forecasting etc.) Train relevant models.

Feature Engineering

Construct more meaningful features out of raw data.

Data Visualization

Create data visualizations to better understand relationship between variables of interest.



Business Understanding

Brainstorm with business users to identify potential use cases for data analytics.

Data Gathering

Identify data sources and pull relevant data to build analytics data-marts.

Data Cleaning

Address the inconsistencies in the data.

Data Exploration

Deep dive into the data to understand what insights can be generated.

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Kitchen Equipment





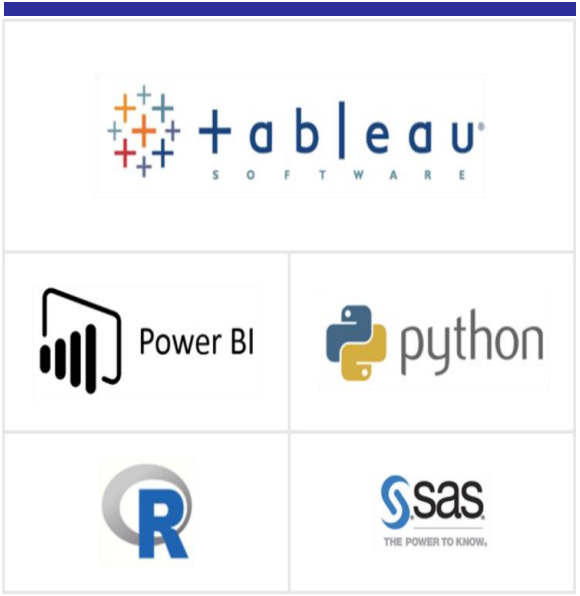
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Important Analytics Tools

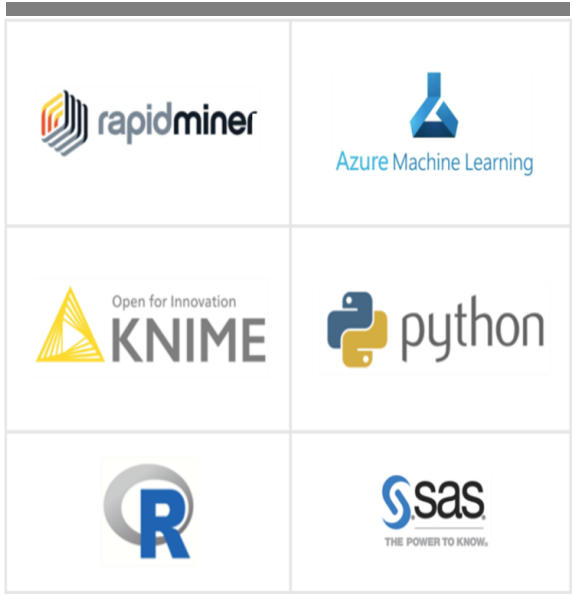
Data Engineering



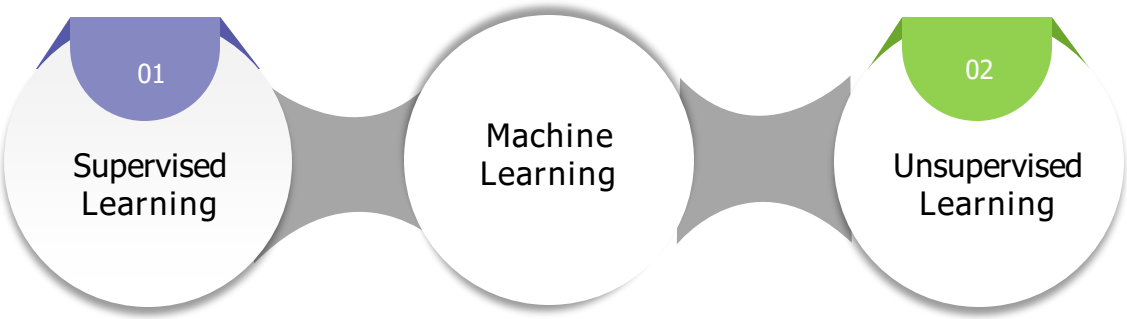
Data Visualization



Data Analytics



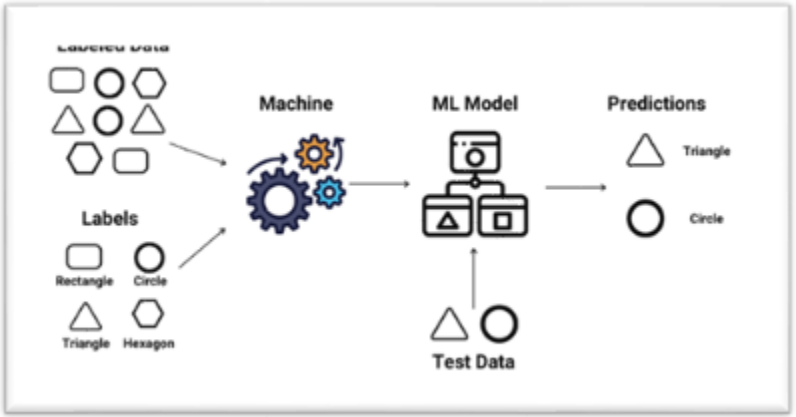
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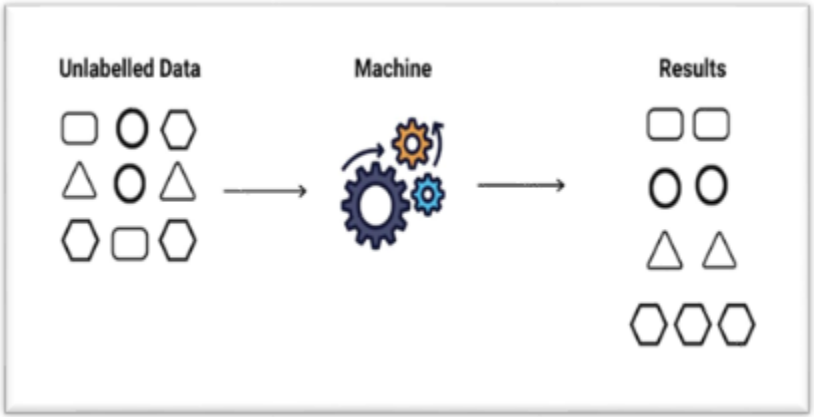
Difference between Supervised Learning and Unsupervised Learning

Supervised Learning



Labeled data required

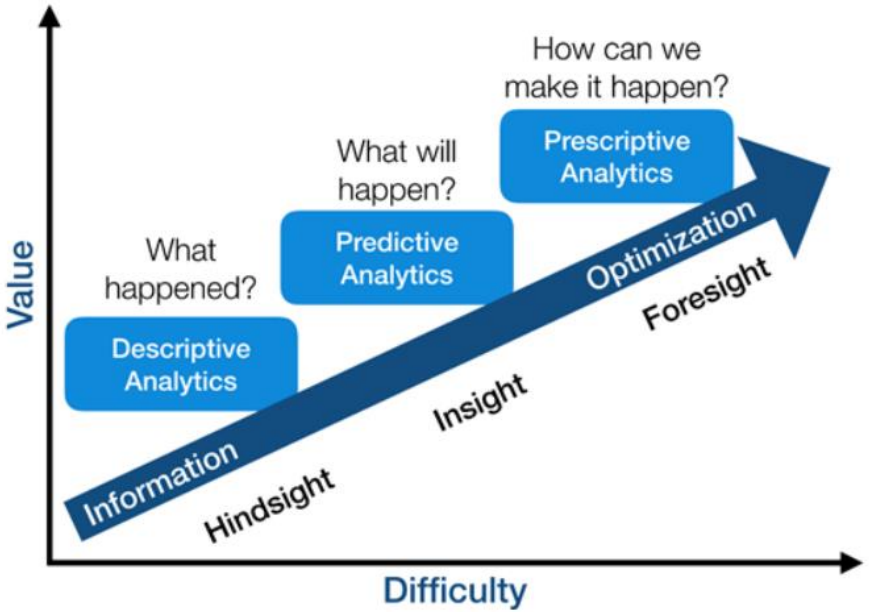
Unsupervised Learning



Labeled data not available

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Machine Learning



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Supervised Machine Learning



N

Supervised Machine Learning

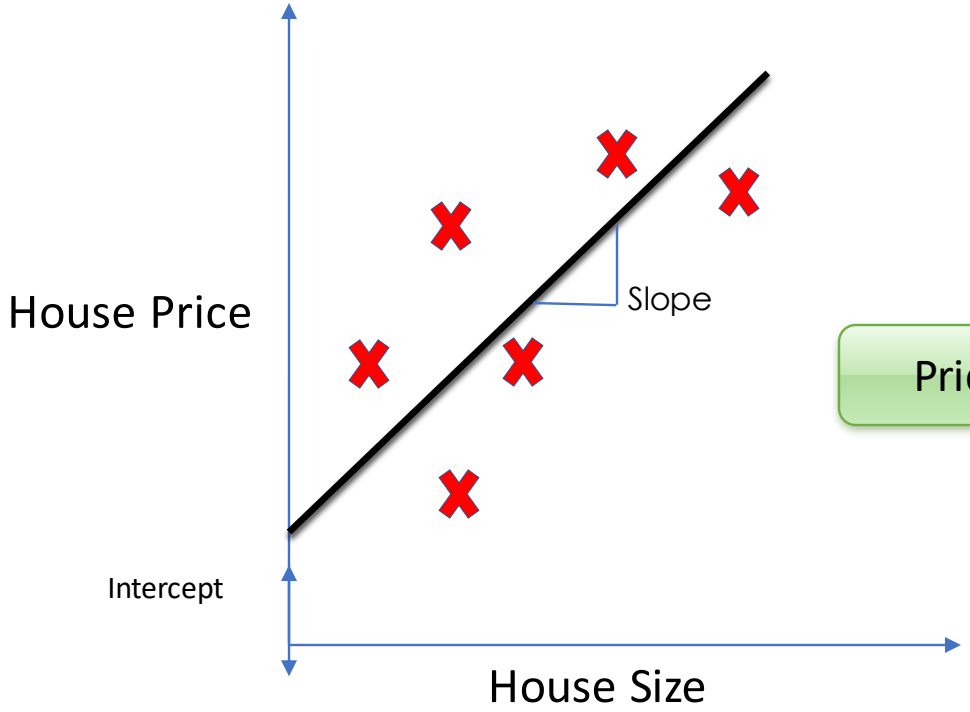
- Linear Regression
- Logistic Regression
- Decision Tree
- Random Forest
- K- Nearest Neighbours
- Support Vector Machine
- Neural Network

Supervised Deep Learning

- Convolutional Neural Network
- Recurrent Neural Network

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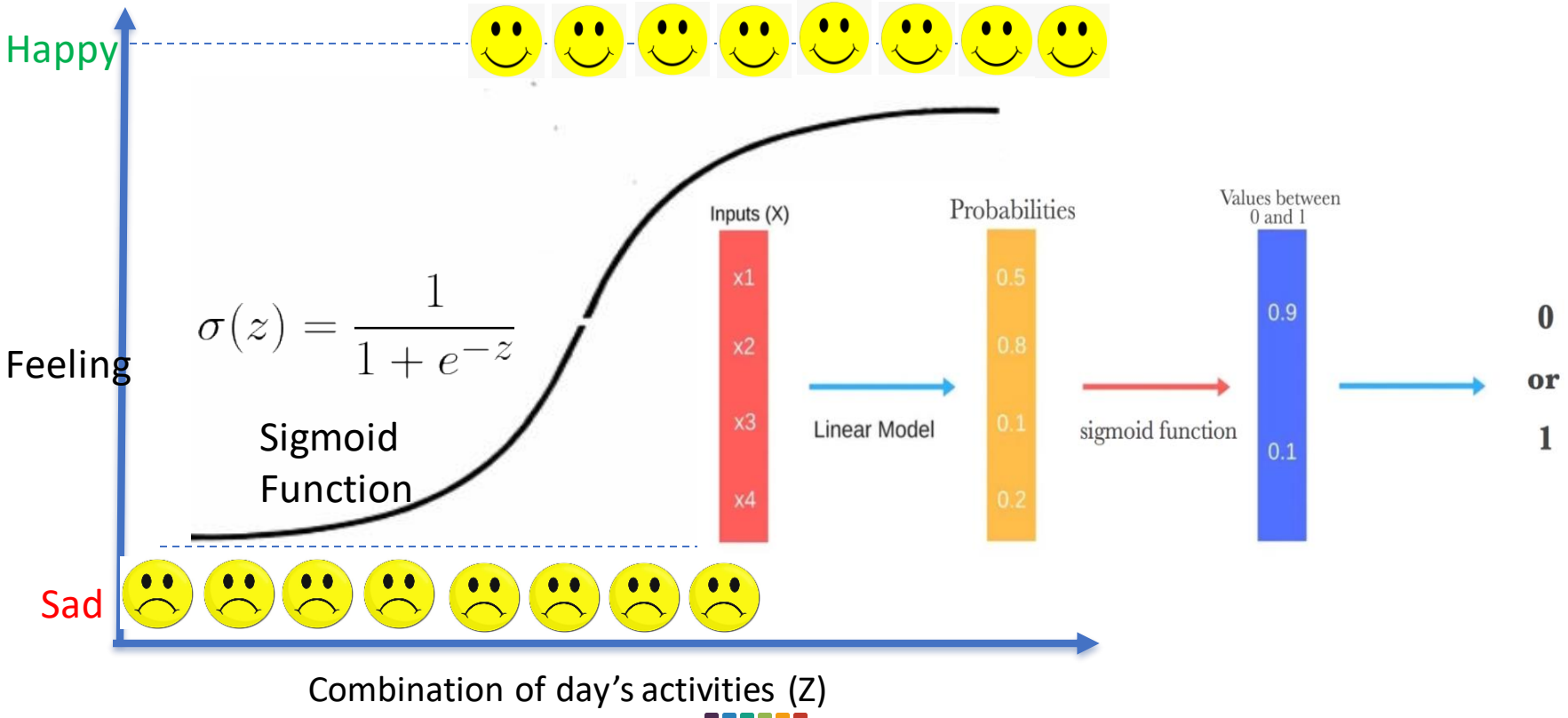
Regression : Linear regression



Price = Intercept + Slope * Size

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Classification : Logistic regression



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Regression/Classification : Decision Tree

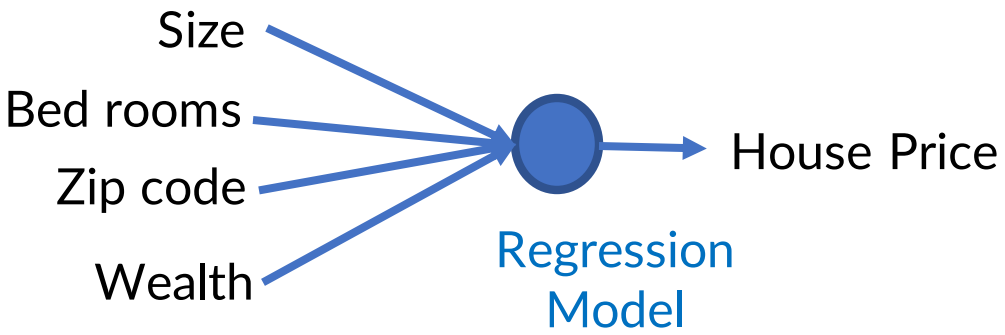


A decision tree is a picture of all the possible courses of action and the consequent possible outcomes.

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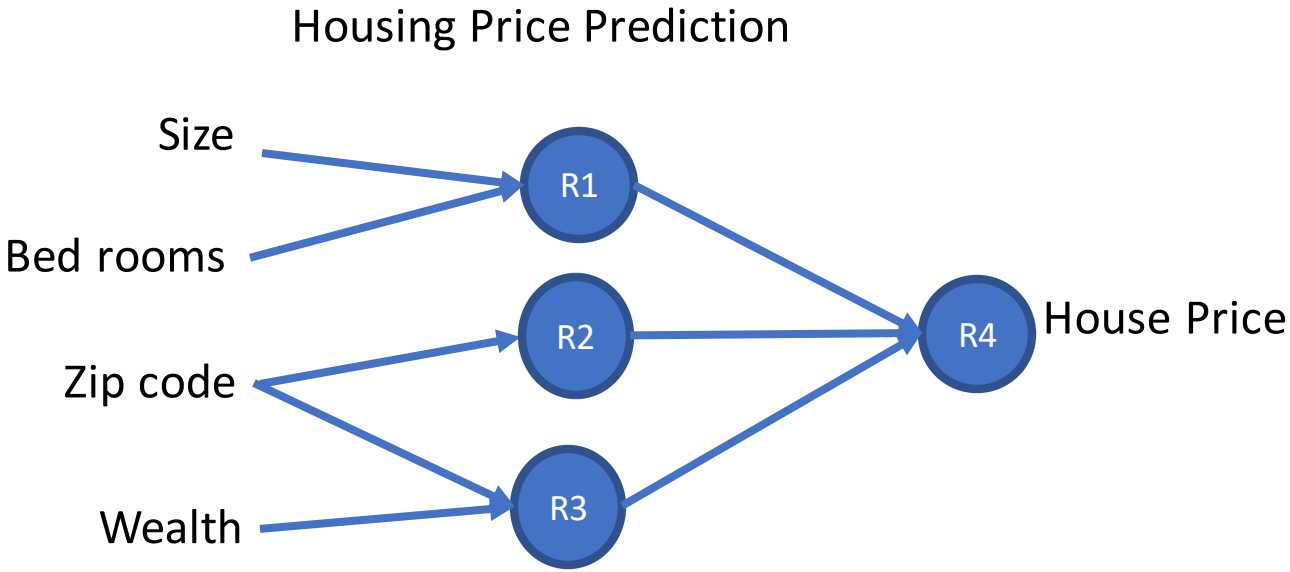
Regression/Classification : Neural Networks

Housing Price Prediction



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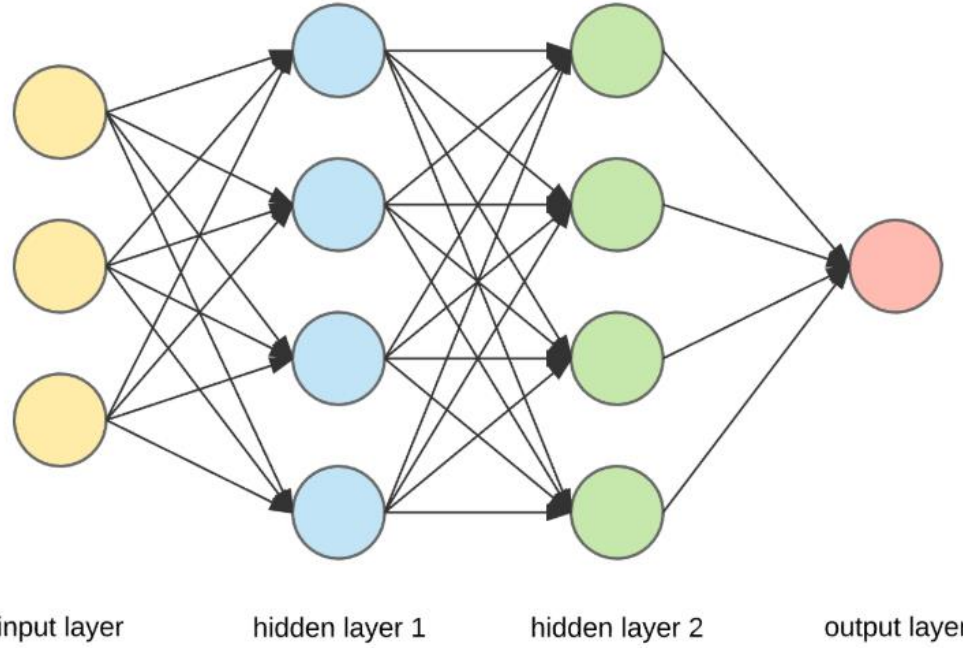
Regression/Classification : Neural Networks



- Sequence of models fitted in a particular pattern
- A large number of relatively uncorrelated models, operating as a committee, may outperform any of the individual constituent models [Wisdom of crowds].

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Regression/Classification : Neural Networks



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Time Series Analysis

Classical Models

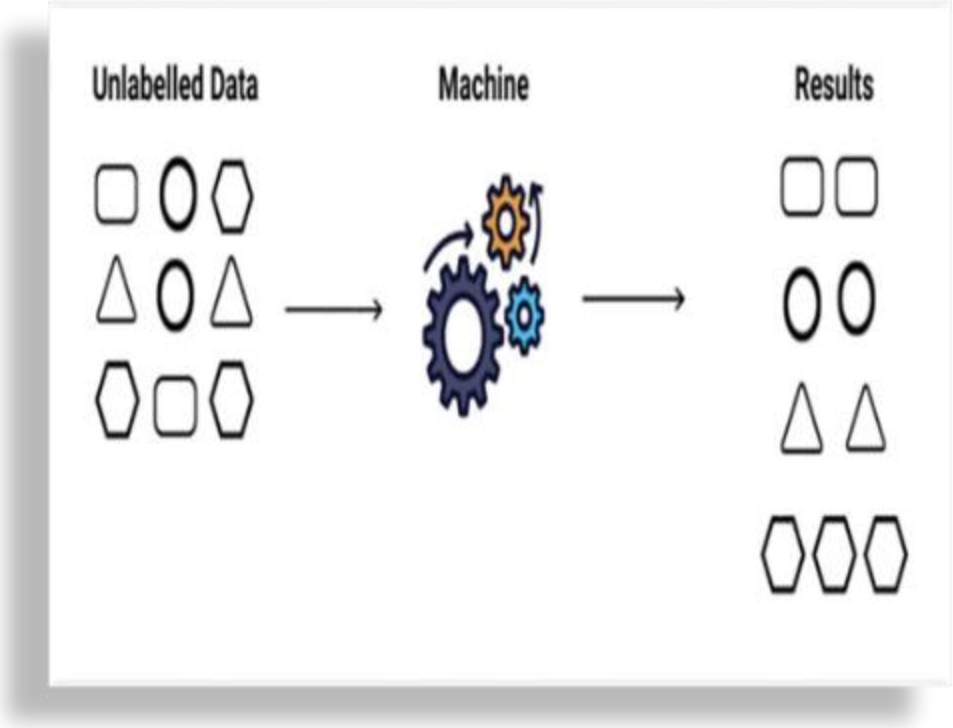
- Random walk
- Moving average
- Simple exponential smoothing
- Holt's exponential smoothing
- Holt-Winter's exponential smoothing
- Autoregressive moving average (ARMA)
- Autoregressive integrated moving average (ARIMA)

New age Models

- Facebook prophet
- Theta
- Recurrent neural network
- Nbeats
- Hybrid

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Unsupervised Learning



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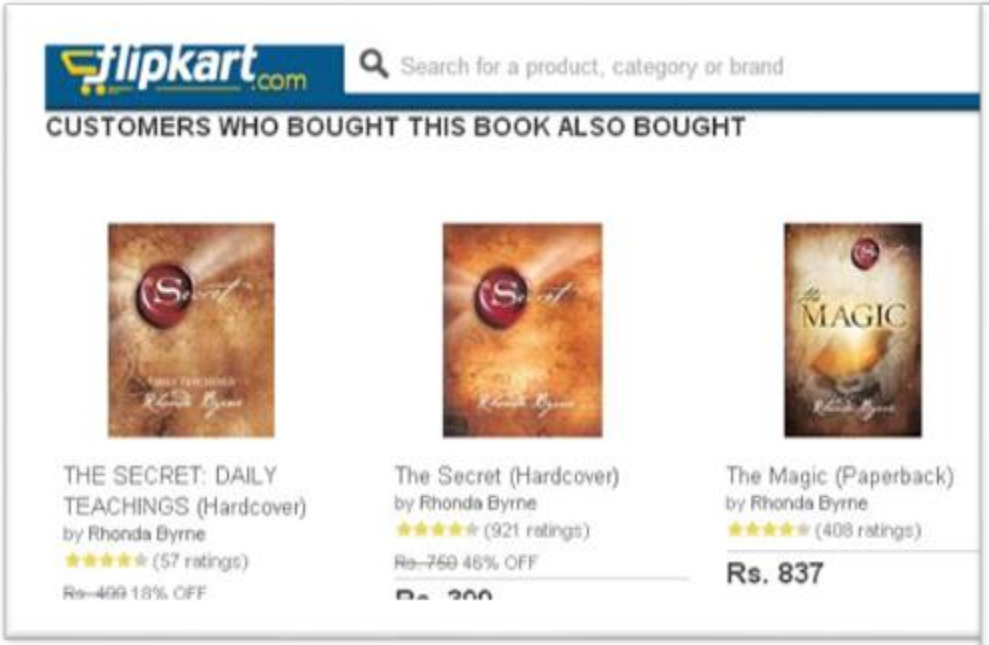
Clustering

- Method of identifying groups in the data
- Types of clustering:
  - Rule based Clustering
  - K Means Clustering
  - Hierarchical Clustering



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Association Rule Mining



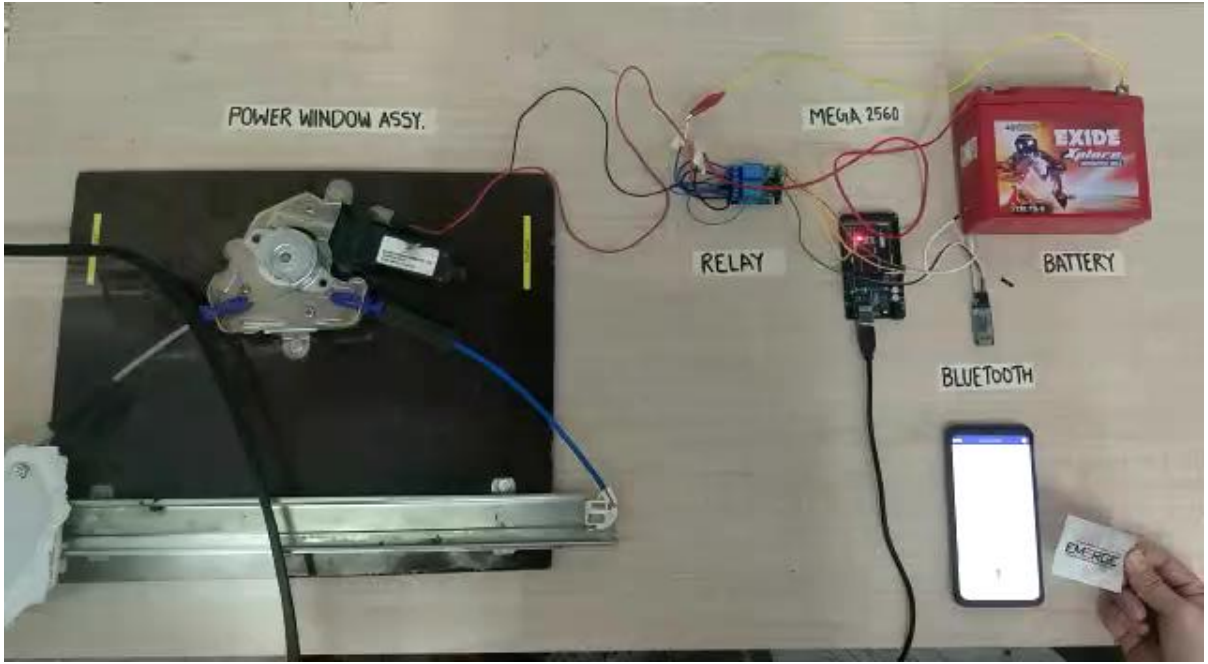


Customers who bought this…also bought…

….. “if this, then that”

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IoT Analytics : Car Window Activation using Voice and Gesture



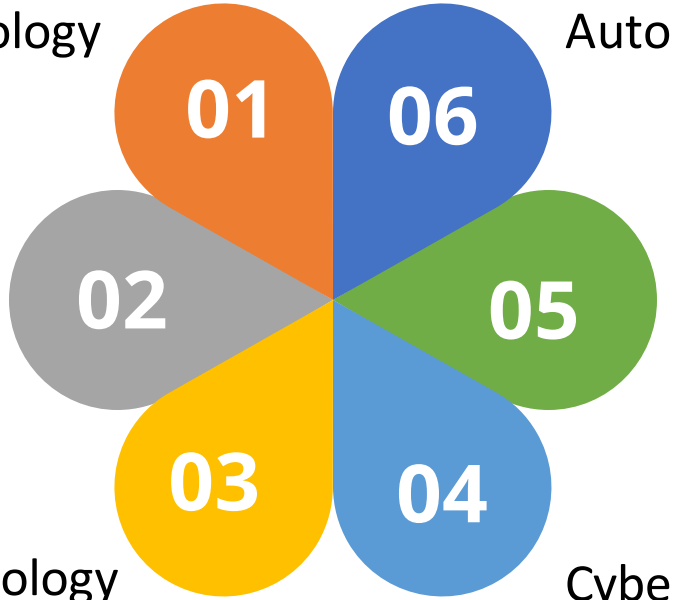
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Our Expertise

Electric Vehicle Technology

Future Mobility Overview

ADAS & Autonomous Technology



Automotive Electronics

IoT, Connectivity & Advanced Computing

Cyber Security for Automotive

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